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Tutorial: 1

Review of Matrix Operations and Derivative for Deep Learning

1. Preliminary Concepts - Basic set Theory and Function

1.4 Practice Problem on Preliminary Concepts

1. $A = \{x: x \in \mathbb{N} \text{ and } x \text{ is even}\}$

$$B = \{x: x \in \mathbb{N} \text{ and } x \text{ is prime}\}$$

$$C = \{x: x \in \mathbb{N} \text{ and } x \text{ is multiple of } 5\}$$

(a) $A \cap B$

It contains natural number that is even prime, as 2 is only even prime.

$$A \cap B = \{2\}$$

(b) $B \cap C$

It contains prime number that is multiple of 5.

$$B \cap C = \{5\}$$

(c) $A \cup B$

It contains even numbers and every

prime number.

$$A \cup B = \{2, 3, 4, 5, 6, 7, 8, 10, 11, \dots\}$$

d) $A \cap (B \cup C)$

This contains natural numbers that are even and either prime or multiple of 5.

Even and prime is 2.

Even and multiple of 5 are multiples of 10.

like 10, 20, 30, ...

$$A \cap (B \cup C) = \{2, 10, 20, 30, \dots\}$$

2. Prove:

$$i) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

To prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$,

we'll show both sides contain same elements by proving following two inclusions.

$$a. A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$$

$$b. (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$$

a. Take an arbitrary element

$$x \in A \cup (B \cap C).$$

By defⁿ of union, $x \in A$ or $x \in (B \cap C)$.

Case 1: If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$.

because $A \subseteq A \cup B$ and $A \subseteq A \cup C$.

Thus, $x \in (A \cup B) \cap (A \cup C)$.

Case 2: If $x \in B \cap C$, then $x \in B$ and $x \in C$.

since $x \in B$, it follows that $x \in A \cup B$.

Similarly, since $x \in C$, it follows that

$x \in A \cup C$. Therefore, $x \in (A \cup B) \cap (A \cup C)$.

Since both case imply that $x \in (A \cup B) \cap (A \cup C)$,

we have satisfied first inclusion,

$$i.e. A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$$

b. Take another arbitrary element

$$x \in (A \cup B) \cap (A \cup C)$$

By defⁿ of intersection, $x \in (A \cup B)$ and

$$x \in (A \cup C)$$

Case 1: If $x \in A$, then $x \in A \cup (B \cap C)$,
because $A \subseteq A \cup (B \cap C)$

Case 2: If $x \notin A$, then since $x \in A \cup B$
and $x \notin A$, it must be that $x \in B$.
Similarly, since $x \in A \cup C$ and $x \notin A$,
it must be that $x \in C$.

Therefore, $x \in (B \cap C)$, so $x \in A \cup (B \cap C)$

Since both cases imply that $x \in A \cup (B \cap C)$,
we have satisfied second inclusion as
well, i.e. $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

As, both inclusions are proven, we can
conclude: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

To prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
we will show that both sides contain same
elements by proving following two inclusions.

a. $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

b. $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

a. Take an arbitrary element x ,
 $x \in A \cap (B \cup C)$

By defⁿ of Intersection, $x \in A$ and $x \in (B \cup C)$

Case 1: If $x \in A$, then $x \in (A \cap B)$ and $x \in B$,
then $x \in (A \cap B)$ and if so,

$x \in (A \cap B) \cup (A \cap C)$

Case 2: If $x \in A$, and $x \notin B$, then $x \in C$ as
 $x \in (B \cup C)$. As $x \in A$ and $x \in C$,
 $x \in (A \cap C)$, and if so,
 $x \in (A \cap B) \cup (A \cap C)$

Since both cases imply that $x \in (A \cap B) \cup (A \cap C)$,
we have satisfied first inclusion, i.e.

$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

b. Take an arbitrary element,
 $x \in (A \cap B) \cup (A \cap C)$

By defⁿ of Union, $x \in (A \cap B)$ or $x \in (A \cap C)$.

~~Case 1: If $x \in A$, then $x \in A$ If $x \in (A \cap B)$,
then $x \in A$ because $A \subseteq (A \cap B)$.~~

~~As $x \in A$, $x \in A$ and $x \in$~~

Case 1: If $x \in (A \cap B)$, then $x \in A$ and $x \in B$
as $A \subseteq (A \cap B)$ and $B \subseteq (A \cap B)$.

Since $B \subseteq (B \cup C)$, $x \in (B \cup C)$.

As $x \in A$ and $x \in (B \cup C)$,

$x \in A \cap (B \cup C)$.

Case 2: If $x \in (A \cap C)$, then $x \in A$ and $x \in C$,
as $A \subseteq (A \cap C)$ and $C \subseteq (A \cap C)$.

Since $C \subseteq (B \cup C)$, $x \in (B \cup C)$.

As $x \in A$ and $x \in (B \cup C)$,

$x \in A \cap (B \cup C)$.

Since both case imply $x \in A \cap (B \cup C)$, we ~~sho~~
have shown that $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

As both inclusions are proven to be true,
we can conclude that:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

3. Define a function: $f: \mathbb{N} \rightarrow \mathbb{N}$ that is:

a) one-to-one but not onto

b) onto but not one-to-one

a) $f(n) = 2n$

where $f: \mathbb{N} \rightarrow \mathbb{N}$ is defined by multiplying
each natural number by 2 is one-to-one
but not onto.

* One-to-one (Injective):

A function is one-to-one if different
inputs maps to different outputs.

Here,

if $f(n_1) = f(n_2)$, then

$2n_1 = 2n_2$, which implies

$n_1 = n_2$. Therefore $f(n) = 2n$ is injective.

* Not Onto (Surjective):

A function is onto if it produces
output every element in co-domain has a

corresponding element in domain. Here, the codomain is $\mathbb{N}$. However, $f(n) = 2n$ only produces even numbers in $\mathbb{N}$, so there is no $n \in \mathbb{N}$ that maps to any odd natural number. Therefore, the function $f(n) = 2n$ is not onto.

The function $f(n) = 2n$ is one-to-one but not onto.

b) $f(n) = \left\lfloor \frac{n}{2} \right\rfloor$

where $f: \mathbb{N} \rightarrow \mathbb{N}$ is defined by dividing every natural number by 2, and rounding up to the nearest integer (floor function).

* Onto (Surjective)

A function is onto if every element in the codomain has at least one corresponding element in the domain. Here, for every any natural number y in codomain $\mathbb{N}$, we can choose $n = 2y$ in the codomain. Then,

$$f(2y) = \left\lfloor \frac{2y}{2} \right\rfloor = \lfloor y \rfloor = y$$

Since every natural number y can be reached, the function is onto.

* Not One-to-One (Injective)

A function is one-to-one if different inputs always map to different output. However, in this function, multiple inputs can produce the same output.

For example, if $n = 1$, then $f(1) = \left\lfloor \frac{1}{2} \right\rfloor = 0$
 if $n = 2$, then $f(2) = \left\lfloor \frac{2}{2} \right\rfloor = 1$

Since, $f(1) = f(2)$ but $1 \neq 2$, the function is not one-to-one.

The function $f(n) = \left\lfloor \frac{n}{2} \right\rfloor$ is onto but not one-to-one.

2. Matrix Operations

2.1. Practice Problem on Basic Matrix Operations.

1.b) Write down 3×4 matrix with 1s on diagonal and 0s elsewhere

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}_{3 \times 4}$$

1.c) Write down 4×3 matrix with 1s on the diagonal and 0s elsewhere

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{4 \times 3}$$

2. Compute AD and DA .

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\rightarrow AD = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}, \quad DA = \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

We can observe $AD \neq DA$.

3. Verify $AB = AC$; $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$

$$\rightarrow AB = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}, \quad AC = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}; \quad AB = AC$$

4. Prove that: $(A.B)^T = B^T.A^T$

$\rightarrow$ To prove $(A.B)^T = B^T.A^T$, we'll show that the element in the i -th row and j -th column of $(A.B)^T$ is identical to the corresponding element in $B^T.A^T$.

1. $(AB)^T$

Let $C = AB$, By defⁿ of matrix multiplication, the element c_{ij} is:

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

By defⁿ of Transpose, the element in the i -th row and j -th column of C^T is the element

from the j -th row and i -th column of C .

let's denote the ij -th element of $(AB)^T$ as $((AB)^T)_{ij}$.

$$((AB)^T)_{ij} = c_{ij} = \sum_{k=1}^n a_{jk} b_{ki} \quad \text{--- (1)}$$

2. $B^T A^T$

let $P = B^T$ with elements $p_{ik} = b_{ki}$

let $Q = A^T$ with elements $q_{kj} = a_{jk}$

The ij -th element of the product $B^T A^T$ is found by multiplying i -th row of B^T by the j -th column of A^T .

$$(B^T A^T)_{ij} = \sum_{k=1}^n p_{ik} q_{kj}$$

Substituting original element a and b :

$$(B^T A^T)_{ij} = \sum_{k=1}^n b_{ki} a_{jk} \quad \text{--- (2)}$$

Since, eqⁿ (1) and (2) are equal ($b_{ki} a_{jk} = a_{jk} b_{ki}$),
Therefore,

$$(AB)^T = B^T A^T$$

5. Compute $A^T A$ and show it is a symmetric square matrix. Also observe sum of the diagonal elements of $A^T A$ is the sum of squares of the elements of A .

$$A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$\rightarrow A^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

$$A^T A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} = \begin{bmatrix} a^2+b^2+c^2 & ad+be+cf \\ da+eb+fc & d^2+e^2+f^2 \end{bmatrix}$$

~~is a~~ $A^T A$ is a symmetric matrix as elements of $(A^T A)_{12} = (A^T A)_{21}$ and $A^T A$ is of size 2×2 .
Therefore, $(A^T A)$ is a symmetric square matrix.

$$\text{sum of diagonals of } (A^T A) = a^2+b^2+c^2+d^2+e^2+f^2$$

$$\text{sum of squares of ele of } A = a^2+b^2+c^2+d^2+e^2+f^2$$

sum of diagonals of $A^T A$ is equal to sum of squares of elements of A .

6. Find the inverse of the matrices:

$$A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$$

$$\rightarrow |A| = 25 - 24 = 1$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix} = \frac{1}{1} \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix} = \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$$|B| = 18 - 12 = 6$$

$$B^{-1} = \frac{1}{|B|} \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2/3 \\ 1/2 & 1/2 \end{bmatrix}$$

7. Verify A and B are inverses.

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}; \quad B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

$\rightarrow$ If A and B are inverses, then

$AB = BA = I_2$ must be true.

$$AB = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 4-3 & -4+4 \\ 3-3 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 4-3 & 1-1 \\ -12+12 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

As $AB = BA = I_{2 \times 2}$,

A and B are inverses of each other.

8. Verify C and D are inverses.

$$C = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}; \quad D = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

$\rightarrow$ To verify C and D are inverses of each other,

we'll prove $CD = DC = I_3$

$$CD = \begin{bmatrix} 3-2 & -4+4 & 1-1 \\ 3-3 & -4+5 & 1-1 \\ 6+6-12 & -8-12+10 & 2+3-4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

$$DC = \begin{bmatrix} 3-4+2 & -3+3 & 4-4 \\ 2-4+2 & -2+3 & 4-4 \\ 3-5+2 & -3+3 & 5-4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

As $CD = DC = I_3$ is proved, we can verify C and D are inverses of each other.

3. System of Linear Equations

3.4 Practice Problem on Solving System of Linear Equations using Matrix Method.

1. System of Two Variables:

$$a. \begin{cases} x + 2y = 4 \\ 3x - 5y = 1 \end{cases}$$

matrix

→ let $A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$ be the representation

of given sys. of linear eq.

$$|A| = \begin{vmatrix} 1 & 2 \\ 3 & -5 \end{vmatrix} = -5 - 6 = -11$$

Replacing first column with vector of result of eqⁿs: $A_x = \begin{bmatrix} 4 & 2 \\ 1 & -5 \end{bmatrix}$

$$|A_x| = \begin{vmatrix} 4 & 2 \\ 1 & -5 \end{vmatrix} = -20 - 2 = -22$$

Replacing second column with vector of result of eqⁿs: $A_y = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$

$$|A_y| = \begin{vmatrix} 1 & 4 \\ 3 & 1 \end{vmatrix} = 1 - 12 = -11$$

$$x = \frac{|A_x|}{|A|} = \frac{-22}{-11} = 2$$

$$y = \frac{|A_y|}{|A|} = \frac{-11}{-11} = 1$$

Solution: $x = 2, y = 1$.

$$b. \begin{cases} 5x + y = 13 \\ 3x + 2y = 5 \end{cases}$$

→ let $B = \begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix}$ be the matrix representation of the given sys. of linear eq.

$$|B| = \begin{vmatrix} 5 & 1 \\ 3 & 2 \end{vmatrix} = 10 - 3 = 7$$

second

Replacing first column with vector of result of eqⁿs

$$B_x = \begin{bmatrix} 13 & 1 \\ 5 & 2 \end{bmatrix}, \quad B_y = \begin{bmatrix} 5 & 13 \\ 3 & 5 \end{bmatrix}$$

$$|B_x| = \begin{vmatrix} 13 & 1 \\ 5 & 2 \end{vmatrix} = 26 - 5 = 21; \quad |B_y| = \begin{vmatrix} 5 & 13 \\ 3 & 5 \end{vmatrix} = 25 - 39 = -14$$

$$x = \frac{|B_x|}{|B|} = \frac{21}{7} = 3; \quad y = \frac{|B_y|}{|B|} = \frac{-14}{7} = -2$$

Solution: $x = 3, y = -2$

$$c. \begin{cases} 3x + 2y = -2 \\ x + 4y = 6 \end{cases}$$

→ let $C = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ be the matrix representation of given sys. of linear eq.

$$|C| = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = 12 - 2 = 10$$

Replacing first & second column of C with results of eqⁿ to obtain: $C_x = \begin{bmatrix} -2 & 2 \\ 6 & 4 \end{bmatrix}$ and $C_y = \begin{bmatrix} 3 & -2 \\ 1 & 6 \end{bmatrix}$

$$|C_x| = -8 - 12 = -20 ; |C_y| = 18 + 2 = 20$$

$$x = \frac{|C_x|}{|C|} = \frac{-20}{10} = -2 ; y = \frac{|C_y|}{|C|} = \frac{20}{10} = 2$$

Solution: $x = -2, y = 2$

$$d. \begin{cases} 2x + 3y = 6 \\ x - y = 1/2 \end{cases}$$

→ let $D = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$ be the matrix representation of given sys. of linear eq.

$$|D| = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = -2 - 3 = -5$$

Replacing first & second column of D with results

$$\text{of eqⁿs to obtain: } D_x = \begin{bmatrix} 6 & 3 \\ 1/2 & -1 \end{bmatrix} \text{ and } D_y = \begin{bmatrix} 2 & 6 \\ 1 & 1/2 \end{bmatrix}$$

$$|D_x| = -6 - 1.5 = -7.5 ; |D_y| = 1 - 6 = -5$$

$$x = \frac{|D_x|}{|D|} = \frac{-7.5}{-5} = +1.5 ; y = \frac{|D_y|}{|D|} = \frac{-5}{-5} = +1$$

Solution: $x = 1.5, y = 1$

$$e. \begin{cases} 4x + 3y = 11 \\ x - 3y = -1 \end{cases}$$

→ let $E = \begin{bmatrix} 4 & 3 \\ 1 & -3 \end{bmatrix}$ be the matrix representation of given sys. of linear eq.

$$|E| = \begin{vmatrix} 4 & 3 \\ 1 & -3 \end{vmatrix} = -12 - 3 = -15$$

Replacing first and second column of E with results & vector of eqⁿ: $E_x = \begin{bmatrix} 11 & 3 \\ -1 & -3 \end{bmatrix} ; E_y = \begin{bmatrix} 4 & 11 \\ 1 & -1 \end{bmatrix}$

$$|E_x| = -33 + 3 = -30 ; |E_y| = -4 - 11 = -15$$

$$x = \frac{|E_x|}{|E|} = \frac{-30}{-15} = 2 ; y = \frac{|E_y|}{|E|} = \frac{-15}{-15} = 1$$

Solution: $x = 2, y = 1$

$$f) \begin{cases} 2x + 3y = 8 \\ 4x + 6y = 20 \end{cases}$$

→ let $F = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$ be the matrix representation of given linear eq.

$$|F| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0$$

The determinant of the matrix representation is 0. ~~How~~ This means that there is no solution to the system of linear eqⁿ. If the lines are either parallel or represent same line.

Looking at resultant vector $[8, 20]$, they are not multiple of one another. Meaning lines are not multiple (scaled), so they are not same.

The system of linear systems represent parallel lines.

2. System with Three Variable

$$a. \begin{cases} x + y - z = 2 \\ x + z = 7 \\ 2x + y + z = 13 \end{cases}$$

→ let $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ represent the matrix representation of given linear eq.

$$A = \begin{vmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -1 - 1(-1) - 1(1) = -1$$

Replacing first, second, third column turn-wise on A with resultant vector.

$$A_x = \begin{bmatrix} 2 & 1 & -1 \\ 7 & 0 & 1 \\ 13 & 1 & 1 \end{bmatrix}; A_y = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 7 & 1 \\ 2 & 13 & 1 \end{bmatrix}; A_z = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 7 \\ 2 & 1 & 13 \end{bmatrix}$$

$$A_x = \begin{vmatrix} 2 & 1 & -1 \\ 7 & 0 & 1 \\ 13 & 1 & 1 \end{vmatrix} = 2(-1) - 1(7-13) - 1(7) = -3$$

$$A_y = \begin{vmatrix} 1 & 2 & -1 \\ 1 & 7 & 1 \\ 2 & 13 & 1 \end{vmatrix} = 1(7-13) - 2(1-2) - 1(13-14) = -6 + 2 + 1 = -3$$

$$A_z = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 0 & 7 \\ 2 & 1 & 13 \end{vmatrix} = 1(-7) - 1(13-14) + 2(1) = -7 + 1 + 2 = -4$$

$$x = \frac{|A_x|}{|A|} = \frac{-3}{-1} = 3, \quad y = \frac{|A_y|}{|A|} = \frac{-3}{-1} = 3, \quad z = \frac{|A_z|}{|A|} = \frac{-4}{-1} = 4$$

$$b. \begin{cases} x + y - z = 2 \\ 3x + y = 7 \\ x + y + 2z = 3 \end{cases}$$

→ let $B = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ be the matrix representation of given sys of linear eq.

$$|B| = \begin{vmatrix} 1 & 1 & -1 \\ 3 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix} = 1(2) - 1(6) - 1(3-1) \\ = 2 - 6 - 2 \\ = -6$$

Replacing 1st, 2nd, 3rd column with resultant vector:

$$B_x = \begin{bmatrix} 2 & 1 & -1 \\ 7 & 1 & 0 \\ 3 & 1 & 2 \end{bmatrix}, B_y = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 7 & 0 \\ 1 & 3 & 2 \end{bmatrix}, B_z = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$$

$$|B_x| = \begin{vmatrix} 2 & 1 & -1 \\ 7 & 1 & 0 \\ 3 & 1 & 2 \end{vmatrix} = 2(2) - 1(14) - 1(7-3) \\ = 4 - 14 - 4 = -14$$

$$|B_y| = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 7 & 0 \\ 1 & 3 & 2 \end{vmatrix} = 1(14) - 2(6) - 1(9-7) \\ = 14 - 12 - 2 = 0$$

$$|B_z| = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 1 & 7 \\ 1 & 1 & 3 \end{vmatrix} = 1(3-7) - 1(9-7) + 2(3-1) \\ = -4 - 2 + 4 = -2$$

$$x = \frac{|B_x|}{|B|} = \frac{-14}{-6} = \frac{7}{3}; y = \frac{|B_y|}{|B|} = \frac{0}{-6} = 0; z = \frac{|B_z|}{|B|} = \frac{-2}{-6} = \frac{1}{3}$$

$$\text{Solution: } x = \frac{7}{3}, y = 0, z = \frac{1}{3}$$

$$c. \begin{cases} -x - 2y + z = 1 \\ 2x + 3y = 2 \\ y - 2z = 0 \end{cases}$$

→ let $C = \begin{bmatrix} -1 & -2 & 1 \\ 2 & 3 & 0 \\ 0 & 1 & -2 \end{bmatrix}$ be the matrix representation of given sys. of linear eqⁿ.

$$|C| = \begin{vmatrix} -1 & -2 & 1 \\ 2 & 3 & 0 \\ 0 & 1 & -2 \end{vmatrix} = -1(-6) + 2(-4) + 1(2) \\ = 6 - 8 + 2 \\ = 0$$

The main matrix determinant is 0.

Therefore, the given sys. of linear eqⁿ has no solution.

$$d. \begin{cases} x + 4y - z = 4 \\ 2x + 5y + 8z = 15 \\ x + 3y - 3z = 1 \end{cases}$$

→ let $D = \begin{bmatrix} 1 & 4 & -1 \\ 2 & 5 & 8 \\ 1 & 3 & -3 \end{bmatrix}$ be the matrix representation of given linear eqⁿ.

$$|D| = \begin{vmatrix} 1 & 4 & -1 \\ 2 & 5 & 8 \\ 1 & 3 & -3 \end{vmatrix} = 16$$

$$|D_x| = \begin{vmatrix} 4 & 4 & -1 \\ 15 & 5 & 8 \\ 1 & 3 & -3 \end{vmatrix} = 16$$

$$|D_y| = \begin{vmatrix} 1 & 4 & -1 \\ 2 & 15 & 8 \\ 1 & 1 & -3 \end{vmatrix} = 16$$

$$|D_z| = \begin{vmatrix} 1 & 4 & 4 \\ 2 & 5 & 15 \\ 1 & 3 & 1 \end{vmatrix} = 16$$

Solution:

$$x = \frac{|D_x|}{|D|} = \frac{16}{16} = 1$$

$$y = \frac{|D_y|}{|D|} = \frac{16}{16} = 1$$

$$z = \frac{|D_z|}{|D|} = \frac{16}{16} = 1$$

$$e. \begin{cases} 5x + 3y + 9z = -1 \\ -2x + 3y - z = -2 \\ -x - 4y + 5z = 1 \end{cases}$$

→ let $E = \begin{bmatrix} 5 & 3 & 9 \\ -2 & 3 & -1 \\ -1 & -4 & 5 \end{bmatrix}$ be the matrix representation of given sys. of linear eqⁿ.

$$|E| = \begin{vmatrix} 5 & 3 & 9 \\ -2 & 3 & -1 \\ -1 & -4 & 5 \end{vmatrix} = 187$$

Solution

$$|E_x| = \begin{vmatrix} -1 & 3 & 9 \\ -2 & 3 & -1 \\ 1 & -4 & 5 \end{vmatrix} = 61 \quad x = \frac{|E_x|}{|E|} = \frac{61}{187}$$

$$|E_y| = \begin{vmatrix} 5 & -1 & 9 \\ -2 & -2 & -1 \\ -1 & 1 & 5 \end{vmatrix} = -92 \quad y = \frac{|E_y|}{|E|} = \frac{-92}{187}$$

$$|E_z| = \begin{vmatrix} 5 & -1 & 9 \\ -2 & -2 & -1 \\ -1 & 1 & 5 \end{vmatrix} = -24 \quad z = \frac{|E_z|}{|E|} = \frac{-24}{187}$$

$$f. \begin{cases} 6x - 5y + 2z = 3 \\ 2x + y - 4z = 5 \\ 3x - 3y + z = -1 \end{cases}$$

→ let $F = \begin{bmatrix} 6 & -5 & 2 \\ 2 & 1 & -4 \\ 3 & -3 & 1 \end{bmatrix}$ be the matrix representation of the given sys. of linear eqⁿ.

$$|F| = \begin{vmatrix} 6 & -5 & 2 \\ 2 & 1 & -4 \\ 3 & -3 & 1 \end{vmatrix} = -14$$

$$F_x = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 1 & -4 \\ -1 & -3 & 1 \end{bmatrix}, |F_x| = \begin{vmatrix} 3 & -5 & 2 \\ 5 & 1 & -4 \\ -1 & -3 & 1 \end{vmatrix} = -56$$

$$F_y = \begin{bmatrix} 6 & 3 & 2 \\ 2 & 5 & -4 \\ 3 & -1 & 1 \end{bmatrix}, |F_y| = \begin{vmatrix} 6 & 3 & 2 \\ 2 & 5 & -4 \\ 3 & -1 & 1 \end{vmatrix} = -70$$

$$F_z = \begin{bmatrix} 6 & -5 & 3 \\ 2 & 1 & 5 \\ 3 & -3 & -1 \end{bmatrix}, |F_z| = \begin{vmatrix} 6 & -5 & 3 \\ 2 & 1 & 5 \\ 3 & -3 & -1 \end{vmatrix} = -28$$

$$x = \frac{|F_x|}{|F|} = \frac{-56}{-14} = 4; \quad y = \frac{|F_y|}{|F|} = \frac{-70}{-14} = 5$$

$$z = \frac{|F_z|}{|F|} = \frac{-28}{-14} = 2$$

Solution: $x = 4, y = 5, z = 2$

3. Bonus Problem

1. Confirm why following sys has no solⁿ.

$$\begin{cases} x + 2y - 3z = -1 \\ x - 3y + z = 1 \\ 2x - y - 3z = 2 \end{cases}$$

→ Let $A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & -3 & 1 \\ 2 & -1 & -3 \end{bmatrix}$ be matrix representation of given sys. of linear eqⁿ.

$$|A| = \begin{vmatrix} 1 & 2 & -3 \\ 1 & -3 & 1 \\ 2 & -1 & -3 \end{vmatrix} = 1(9+1) - 2(-3-2) - 3(-1+6) \\ = 10 + 10 - 15 \\ = 5$$

$$A_x = \begin{bmatrix} -1 & 2 & -3 \\ 1 & -3 & 1 \\ 2 & -1 & -3 \end{bmatrix}; |A_x| = \begin{vmatrix} -1 & 2 & -3 \\ 1 & -3 & 1 \\ 2 & -1 & -3 \end{vmatrix} = -15$$

$$A_y = \begin{bmatrix} 1 & -1 & -3 \\ 1 & 1 & 1 \\ 2 & 2 & -3 \end{bmatrix}; |A_y| = \begin{vmatrix} 1 & -1 & -3 \\ 1 & 1 & 1 \\ 2 & 2 & -3 \end{vmatrix} = -10$$

$$A_z = \begin{bmatrix} 1 & 2 & -1 \\ 1 & -3 & 1 \\ 2 & -1 & 2 \end{bmatrix}; |A_z| = \begin{vmatrix} 1 & 2 & -1 \\ 1 & -3 & 1 \\ 2 & -1 & 2 \end{vmatrix} = -10$$

$$x = \frac{|A_x|}{|A|} = \frac{-15}{5} = -3, y = \frac{|A_y|}{|A|} = \frac{-10}{5} = -2$$

$$z = \frac{|A_z|}{|A|} = \frac{-10}{5} = -2$$

Solution:

$$x = -3, -2, -2$$

The given system of linear eqⁿ actually have an solution.

2. Confirm why following sys has no solⁿ.

$$\begin{cases} 2x - 2y + 3z = 6 \\ 4x - 3y + 2z = 0 \\ -2x + 3y - 7z = 1 \end{cases}$$

→ Let $B = \begin{bmatrix} 2 & -2 & 3 \\ 4 & -3 & 2 \\ -2 & 3 & -7 \end{bmatrix}$ be matrix representation of given sys of linear eq.

$$|B| = \begin{vmatrix} 2 & -2 & 3 \\ 4 & -3 & 2 \\ -2 & 3 & -7 \end{vmatrix} = 2(21-6) + 2(-28+4) + 3(12-6) \\ = 30 - 48 + 18 \\ = 0$$

Since, the main matrix determinant is 0,

The given sys. of linear eqⁿ has no ^{unique} solution.

To find if the sys. has no solⁿ or infinitely many solⁿ, solving using gauss-jordan elimination.

Augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & -2 & 3 & 6 \\ 4 & -3 & 2 & 0 \\ -2 & 3 & -7 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 / 2$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 4 & -3 & 2 & 0 \\ -2 & 3 & -7 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 4R_1 \text{ and } R_3 \rightarrow R_3 + 2R_1$$

$$\rightarrow 1 \quad \sim \begin{bmatrix} 1 & -1 & 3/2 & : & 3 \\ 0 & 1 & -4 & : & -12 \\ 0 & 1 & -4 & : & 7 \end{bmatrix}$$

$$R_1 \rightarrow R_1 + R_2 \text{ and } R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -5/2 & : & -9 \\ 0 & 1 & -4 & : & -12 \\ 0 & 0 & 0 & : & 19 \end{bmatrix}$$

Backward substitution:

$$0 \cdot z = 19$$

$0 = 19$ which is false

of any value. So, there is no point at the the three lines meet. Meaning lines are parallel. Hence, sys. has no solution.

3. Confirm why sys has infinitely many solⁿ.

$$\begin{cases} x + 2y - z = 1 \\ 2x + 7y + 4z = 11 \\ x + 3y + z = 4 \end{cases}$$

$$\rightarrow \text{let } C = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & 4 \\ 1 & 3 & 1 \end{bmatrix} \quad \text{Augmented matrix:}$$

$$\begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 2 & 7 & 4 & : & 11 \\ 1 & 3 & 1 & : & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 0 & 3 & 6 & : & 9 \\ 0 & 1 & 2 & : & 3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 / 3 \text{ then}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 0 & 1 & 2 & : & 3 \\ 0 & 1 & 2 & : & 3 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2 \text{ and } R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -5 & : & -5 \\ 0 & 1 & 2 & : & 3 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Backward substitution:

$$0 \cdot z = 0$$

$0 = 0$ which is true.

For any value of z , the eqⁿ remains true, so, the ~~lines are over all~~ eqⁿs represent same line. Hence, sys. has infinitely many solutions.

4. Introduction to Derivative in ML

4.1.1. Practice Problem: let's take some derivative
Problem 1: Find $f'(x)$ for following:

$$1. f(x) = x + \sqrt{x}$$

$$\rightarrow f'(x) = \frac{d}{dx}(x + \sqrt{x}) = \frac{dx}{dx} + \frac{d\sqrt{x}}{dx}$$

$$= 1 + \frac{dx^{1/2}}{dx}$$

$$= 1 + \frac{1}{2x^{1/2}}$$

$$= 1 + \frac{1}{2\sqrt{x}}$$

$$2. f(x) = 2 + \frac{1}{x} + \frac{1}{x^2}$$

$$\rightarrow f'(x) = \frac{d}{dx}\left(2 + \frac{1}{x} + \frac{1}{x^2}\right) = \frac{d2}{dx} + \frac{dx^{-1}}{dx} + \frac{dx^{-2}}{dx}$$

$$= -1 \cdot x^{-2} + (-2)x^{-3}$$

$$= -\frac{1}{x^2} - \frac{2}{x^3}$$

$$3. f(x) = \sqrt{x} \cos x$$

$$\rightarrow f'(x) = \frac{d}{dx}(\sqrt{x} \cos x) = \sqrt{x} \frac{d\cos x}{dx} + \cos x \frac{d\sqrt{x}}{dx}$$

$$= \sqrt{x}(-\sin x) + \cos x \cdot \frac{1}{2\sqrt{x}}$$

$$= \frac{\cos x}{2\sqrt{x}} - \sin x \cdot \sqrt{x}$$

$$4. f(x) = x^3 e^x$$

$$\rightarrow f'(x) = \frac{d}{dx}(x^3 e^x) = x^3 \frac{de^x}{dx} + e^x \frac{dx^3}{dx}$$

$$= x^3 e^x + e^x 3x^2$$

$$= x^2 e^x (x + 3)$$

$$5. x^3 / e^x$$

$$\rightarrow f'(x) = \frac{d}{dx}\left(\frac{x^3}{e^x}\right) = \frac{e^x \frac{dx^3}{dx} - x^3 \frac{de^x}{dx}}{(e^x)^2}$$

$$= \frac{3x^2 e^x - x^3 e^x}{e^{2x}}$$

$$= \frac{3x^2 - x^3}{e^x}$$

$$6. f(x) = (3x^2 + 2x + 1)^5$$

$$\rightarrow f'(x) = \frac{d}{dx} (3x^2 + 2x + 1)^5$$

$$= \frac{d(3x^2 + 2x + 1)^5}{d(3x^2 + 2x + 1)} \cdot \frac{d(3x^2 + 2x + 1)}{dx}$$

$$= 5(3x^2 + 2x + 1)^4 \left[\frac{d3x^2}{dx} + \frac{d2x}{dx} + \frac{d1}{dx} \right]$$

$$= 5(3x^2 + 2x + 1)^4 (6x + 2)$$

$$= 5(6x + 2)(3x^2 + 2x + 1)^4$$

$$7. f(x) = (\sin x)^4$$

$$\rightarrow f'(x) = \frac{d(\sin x)^4}{dx}$$

$$= \frac{d(\sin x)^4}{d(\sin x)} \cdot \frac{d(\sin x)}{dx}$$

$$= 4(\sin x)^3 \cdot \cos x$$

$$= 4 \sin^3 x \cdot \cos x$$

$$8. f(x) = e^{(3x^2 + 2x)}$$

$$\rightarrow f'(x) = \frac{d e^{(3x^2 + 2x)}}{dx}$$

$$= \frac{d e^{(3x^2 + 2x)}}{d(3x^2 + 2x)} \cdot \frac{d(3x^2 + 2x)}{dx}$$

$$= e^{(3x^2 + 2x)} \cdot (6x + 2)$$

$$= (6x + 2) e^{(3x^2 + 2x)}$$

$$9. f(x) = \ln(5x^2 + 3x)$$

$$\rightarrow f'(x) = \frac{d \ln(5x^2 + 3x)}{dx}$$

$$= \frac{d \ln(5x^2 + 3x)}{d(5x^2 + 3x)} \cdot \frac{d(5x^2 + 3x)}{dx}$$

$$= \frac{1}{(5x^2 + 3x)} \cdot (10x + 3)$$

$$= \frac{10x + 3}{5x^2 + 3x}$$

$$10. f(x) = \sqrt{x^3 + 4x + 1}$$

$$\rightarrow f'(x) = \frac{d \sqrt{x^3 + 4x + 1}}{dx}$$

$$= \frac{d(x^3 + 4x + 1)^{1/2}}{d(x^3 + 4x + 1)} \cdot \frac{d(x^3 + 4x + 1)}{dx}$$

$$= \frac{1}{2\sqrt{x^3 + 4x + 1}} \cdot (3x^2 + 4)$$

$$= \frac{3x^2 + 4}{2\sqrt{x^3 + 4x + 1}}$$

Problem 2:

$g(x) = x^2 - 3x + 2$ and $f(x)$ is differentiable

$\cdot f(0) = 3 ; f'(0) = -1$

$\cdot f(1) = 5 ; f'(1) = 0$

$\cdot f(2) = -2 ; f'(2) = 3$

$\cdot f(3) = 6 ; f'(3) = 1$

1. Find $\frac{d}{dx} [f(x) \cdot g(x)]$ at $x = 1$.

2. Find $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right]$ at $x = 0$

3. Find $\frac{d}{dx} f(g(x))$ at $x = 0$.

$$g'(x) = \frac{d}{dx} (x^2 - 3x + 2) = 2x - 3$$

$$\xrightarrow{1} \frac{d}{dx} [f(x) \cdot g(x)] = f(x) \cdot g'(x) + g(x) f'(x)$$

$$\begin{aligned} \text{at } x = 1 ; &= f(1)g'(1) + g(1)f'(1) \\ &= 5 \cdot (2 \cdot 1 - 3) + (1^2 - 3 \cdot 1 + 2) \cdot 0 \\ &= 5 \cdot (-1) \\ &= -5 \end{aligned}$$

$$\xrightarrow{2} \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$\text{at } x = 0 ; = \frac{f'(0) \cdot g(0) - f(0) \cdot g'(0)}{(g(0))^2}$$

$$= \frac{-1 \cdot (0^2 - 3 \cdot 0 + 2) - 3 \cdot (-3)}{(2)^2}$$

$$= \frac{-2 + 9}{4}$$

$$= \frac{7}{4}$$

$$\xrightarrow{3} g(0) = 0^2 - 3 \cdot 0 + 2 = 2$$

$$\frac{d}{dx} f(g(x)) = \frac{df(g(x))}{dg(x)} \cdot \frac{dg(x)}{dx}$$

$$= f'(g(x)) \cdot g'(x)$$

$$\text{at } x = 0 ; = f'(g(0)) \cdot g'(0)$$

$$= f'(2) \cdot (2 \cdot 0 - 3)$$

$$= 3 \cdot (-3)$$

$$= -9$$

4.2.1. Practice Problem on Maxima and Minima using derivative.

Problem 1: For the following functions, find the maximum or minimum of function at given interval using critical points, if exist:

1. $g(x) = x^2 - 4x + 5$ at $[1, 4]$

$$\rightarrow g'(x) = \frac{d}{dx}(x^2 - 4x + 5) = 2x - 4$$

$$g'(x) = 0$$

$$2x - 4 = 0$$

$$2x = 4$$

$$x = 2$$

critical point is $x = 2$.

$$g(1) = 1^2 - 4 \cdot 1 + 5 = 2$$

$$g(2) = 2^2 - 4 \cdot 2 + 5 = 3$$

$$g(4) = 4^2 - 4 \cdot 4 + 5 = 5$$

maximum is 5 at $x = 4$

minimum is 2 at $x = 1$

2. $h(x) = \frac{x^2}{2} - 2x$ at $[-2, 3]$

→ 1. Finding derivative: $h'(x) = \frac{d}{dx} \left(\frac{x^2}{2} - 2x \right)$

$$= x - 2$$

2. Finding critical points by equating with 0

$$\Rightarrow x - 2 = 0$$

So, critical point is : $x = 2$ is critical

3. Evaluate $h(x)$ at endpoint & critical points:

$$h(-2) = \frac{(-2)^2}{2} - 2(-2) = 2 + 4 = 6$$

$$h(2) = \frac{2^2}{2} - 2 \cdot 2 = 2 - 4 = -2$$

$$h(3) = \frac{3^2}{2} - 2 \cdot 3 = \frac{9}{2} - 6 = -\frac{3}{2}$$

4. Maximum of $h(x)$ is 6 at $x = -2$

Minimum of $h(x)$ is -2 at $x = 2$

3. $p(x) = 3x^3 - 9x^2 + 7$ at $[-1, 2]$

→ Finding derivative $\Rightarrow p'(x) = \frac{d}{dx}(3x^3 - 9x^2 + 7)$

$$= 9x^2 - 18x$$

Finding critical points by $\Rightarrow 9x^2 - 18x = 0$

equating with 0

$$x(x - 2) = 0$$

$\therefore$ critical points of $p(x)$: $x = 0, 2$

Evaluating $p(x)$ at endpoints of interval & critical points:

$$p(-1) = 3(-1)^3 - 9(-1)^2 + 7 = -3 - 9 + 7 = -5$$

$$p(0) = 3(0)^3 - 9(0) + 7 = 7$$

$$p(2) = 3(2)^3 - 9(2)^2 + 7 = -5$$

Maximum of $p(x)$ is 7 at $x = 0$.

Minimum of $p(x)$ is -5 at $x = -1, 2$

4. $f(x) = x^2 - 4x + 6$ at $[0, 3]$

→ Finding derivative $\Rightarrow f'(x) = \frac{d}{dx}(x^2 - 4x + 6)$

$$= 2x - 4$$

Equating with 0 to find $\Rightarrow 2x - 4 = 0$

critical points

$$x = 2$$

$\therefore$ critical point of $f(x)$ is : $x = 2$

Evaluating $f(x)$ at endpoints of interval and critical points:

$$f(0) = 0^2 - 4(0) + 6 = 6$$

$$f(2) = 2^2 - 4(2) + 6 = 2$$

$$f(3) = 3^2 - 4(3) + 6 = 3$$

Maximum of $f(x)$ is 6 at $x = 0$

Minimum of $f(x)$ is 2 at $x = 2$

5. $q(x) = -2x^3 + 3x^2 + 12x$ at $[-1, 2]$

→ Finding derivative $\Rightarrow q'(x) = \frac{d}{dx} (-2x^3 + 3x^2 + 12x)$

$$= -6x^2 + 6x + 12$$

Finding critical points by $\Rightarrow -6x^2 + 6x + 12 = 0$
equating with 0 $x^2 - x - 2 = 0$

$$(x-2)(x+1) = 0$$

$$x = 2, -1$$

Evaluating $f(x)$ at end points:

$$\begin{aligned} f(-1) &= -2(-1)^3 + 3(-1)^2 + (-1) \cdot 12 \\ &= 2 + 3 - 12 \\ &= -7 \end{aligned}$$

$$\begin{aligned} f(2) &= -2 \cdot 2^3 + 3 \cdot 2^2 + 12 \cdot 2 \\ &= -16 + 12 + 24 \\ &= 20 \end{aligned}$$

Maximum of $f(x)$ is 20 at $x = 2$

Minimum of $f(x)$ is -7 at $x = -1$.

4.3.3 Practice Problem on Gradients

1. Evaluate the gradients of functions at given points:

1. $f(x, y) = x^2 + y^2$ at

(a) $(0, 0)$ (b) $(1, 3)$ (c) $(-1, -5)$

2. Calculating partial derivatives of $f(x, y)$:

$$\nabla f(x, y) = \left[\frac{\partial f(x, y)}{\partial x}, \frac{\partial f(x, y)}{\partial y} \right]$$

$$= \left[\frac{\partial (x^2 + y^2)}{\partial x}, \frac{\partial (x^2 + y^2)}{\partial y} \right]$$

$$= [2x, 2y]$$

(a) At $(0, 0)$:

$$\begin{aligned} \nabla f(0, 0) &= [2 \cdot 0, 2 \cdot 0] \\ &= (0, 0) \end{aligned}$$

(b) At $(1, 3)$:

$$\begin{aligned} \nabla f(1, 3) &= [2 \cdot 1, 2 \cdot 3] \\ &= (2, 6) \end{aligned}$$

(c) At $(-1, -5)$:

$$\begin{aligned} \nabla f(-1, -5) &= (2(-1), 2(-5)) \\ &= (-2, -10) \end{aligned}$$

$$2. f(x, y, z) = x^3z - 2y^2x + 5z \text{ at:}$$

$$(a) (1, 1, -4) \quad (b) (0, 1, 0) \quad (c) (-3, -2, 1)$$

2 → Calculating partial derivative of $f(x, y, z)$:

$$\nabla f(x, y, z) = \left[\frac{\partial f(x, y, z)}{\partial x}, \frac{\partial f(x, y, z)}{\partial y}, \frac{\partial f(x, y, z)}{\partial z} \right]$$

$$\frac{\partial f(x, y, z)}{\partial x} = \frac{\partial (x^3z - 2y^2x + 5z)}{\partial x}$$

$$= 3x^2z - 2y^2$$

$$\frac{\partial f(x, y, z)}{\partial y} = \frac{\partial (x^3z - 2y^2x + 5z)}{\partial y}$$

$$= -4xy$$

$$\frac{\partial f(x, y, z)}{\partial z} = \frac{\partial (x^3z - 2y^2x + 5z)}{\partial z}$$

$$= x^3 + 5$$

$$\nabla f(x, y, z) = [3x^2z - 2y^2, -4xy, x^3 + 5]$$

$$(a) \text{ At } (1, 1, -4) : \nabla f(1, 1, -4) =$$

$$[3 \cdot 1^2 \cdot (-4) - 2 \cdot 1^2, -4 \cdot 1 \cdot 1, 1^3 + 5]$$

$$= [-14, -4, 6]$$

$$(b) \text{ At } (0, 1, 0) : \nabla f(0, 1, 0) =$$

$$[3 \cdot 0^2 \cdot 0 - 2 \cdot 1^2, -4 \cdot 0 \cdot 1, 0^3 + 5]$$

$$= [-2, 0, 5]$$

$$(c) \text{ At } (-3, -2, 1) : \nabla f(-3, -2, 1) =$$

$$[3(-3)^2 \cdot 1 - 2(-2)^2, -4(-3)(-2), (-3)^3 + 5]$$

$$= [19, -24, -22]$$

3. ~~Solved in~~ Solution given

4.3.3.2 Problem: Using Gradient to Evaluate Maxima and Minima of a Multi Variate Function over a specific interval.

For the given functions, evaluate the maximum and minimum values over the specified interval:

$$1. m(x, y) = -x^2 - y^2 + 4x + 6y \text{ at } x: [0, 4], y: [0, 5]$$

$$\rightarrow \text{Gradient of } m(x, y) = \nabla m(x, y) = \left[\frac{\partial m}{\partial x}, \frac{\partial m}{\partial y} \right]$$

Calculating partial derivatives:

$$\frac{\partial m}{\partial x} = \frac{\partial (-x^2 - y^2 + 4x + 6y)}{\partial x} = -2x + 4$$

$$\frac{\partial m}{\partial y} = \frac{\partial (-x^2 - y^2 + 4x + 6y)}{\partial y} = -2y + 6$$

$$\text{Gradient } \nabla m(x, y) = [-2x + 4, -2y + 6]$$

Finding critical points by equating with 0:

$$-2x + 4 = 0 \Rightarrow x = 2$$

$$-2y + 6 = 0 \Rightarrow y = 3$$

Evaluating $m(x, y)$ at critical points and end points:

$$\text{At } (2, 3): m(2, 3) = -2^2 - 3^2 + 4 \cdot 2 + 6 \cdot 3 = 13$$

$$\text{At } (0, 0): m(0, 0) = -0^2 - 0^2 + 4 \cdot 0 + 6 \cdot 0 = 0$$

$$\text{At } (0, 5): m(0, 5) = -0^2 - 5^2 + 4 \cdot 0 + 6 \cdot 5 = 5$$

$$\text{At } (4, 0): m(4, 0) = -4^2 - 0^2 + 4 \cdot 4 + 6 \cdot 0 = 0$$

$$\text{At } (4, 5): m(4, 5) = -4^2 - 5^2 + 4 \cdot 5 + 6 \cdot 5 = 9$$

Max: 13 at (2, 3) and Min: 0 at (0, 0) & (4, 0)

$$2. g(x, y) = xy + x^2 - 4y + 6 \text{ at } x: [-2, 2], y: [1, 3]$$

$$\rightarrow \text{Gradient of } g(x, y) = \nabla g(x, y) = \left[\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y} \right]$$

Calculating partial derivatives:

$$\frac{\partial g}{\partial x} = \frac{\partial (xy + x^2 - 4y + 6)}{\partial x} = y + 2x$$

$$\frac{\partial g}{\partial y} = \frac{\partial (xy + x^2 - 4y + 6)}{\partial y} = x - 4$$

$$\text{Gradient } \nabla g(x, y) = (2x + y, x - 4)$$

Finding critical points: ~~2x~~ $x - 4 = 0 \Rightarrow x = 4$

by equating with 0 $2x + y = 0 \Rightarrow y = -8$

Critical point $(4, -8)$ lies outside the interval, so evaluating critical points $g(x, y)$ on endpoints:

$$\text{At } (-2, 1): g(-2, 1) = -2 \cdot 1 + (-2)^2 - 4 \cdot 1 + 6 = 4$$

$$\text{At } (-2, 3): g(-2, 3) = -2 \cdot 3 + (-2)^2 - 4 \cdot 3 + 6 = -8$$

$$\text{At } (2, 1): g(2, 1) = 2 \cdot 1 + 2^2 - 4 \cdot 1 + 6 = 8$$

$$\text{At } (2, 3): g(2, 3) = 2 \cdot 3 + 2^2 - 4 \cdot 3 + 6 = 4$$

Maximum of $g(x, y) = 8$ at (2, 1)

Minimum of $g(x, y) = -8$ at (-2, 3)

$$3. p(x,y) = x^3 - 3xy + 4 \text{ at } x: [0,2], y: [0,3]$$

$$\rightarrow \text{Gradient of } p(x,y) = \nabla p(x,y) = \left[\frac{\partial p}{\partial x}, \frac{\partial p}{\partial y} \right]$$

Calculating partial derivatives:

$$\frac{\partial p}{\partial x} = \frac{\partial (x^3 - 3xy + 4)}{\partial x} = 3x^2 - 3y$$

$$\frac{\partial p}{\partial y} = \frac{\partial (x^3 - 3xy + 4)}{\partial y} = -3x$$

$$\text{Gradient } \nabla p(x,y) = [3x^2 - 3y, -3x]$$

Finding critical points by equating with 0:

$$\Rightarrow -3x = 0 \Rightarrow x = 0$$

$$\Rightarrow 3x^2 - 3y = 0 \Rightarrow y = 0$$

Since, critical point (0,0) lies at end point,

evaluating $p(x,y)$ in end points:

$$\text{At } (0,0): p(0,0) = 0^3 - 3 \cdot 0 \cdot 0 + 4 = 4$$

$$\text{At } (0,3): p(0,3) = 0^3 - 3 \cdot 0 \cdot 3 + 4 = 4$$

$$\text{At } (2,0): p(2,0) = 2^3 - 3 \cdot 2 \cdot 0 + 4 = 12$$

$$\text{At } (2,3): p(2,3) = 2^3 - 3 \cdot 2 \cdot 3 + 4 = -6$$

$$\text{Maximum of } p(x,y) = 12 \text{ at } (2,0)$$

$$\text{Minimum of } p(x,y) = -6 \text{ at } (2,3)$$

$$4. h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20 \text{ at } x: [1,5], y: [0,4]$$

$$\rightarrow \text{Gradient of } h(x,y) = \nabla h(x,y) = \left[\frac{\partial h}{\partial x}, \frac{\partial h}{\partial y} \right]$$

Calculating partial derivatives:

$$\frac{\partial h}{\partial x} = \frac{\partial (2x^2 + 3y^2 - 12x - 18y + 20)}{\partial x} = 4x - 12$$

$$\frac{\partial h}{\partial y} = \frac{\partial (2x^2 + 3y^2 - 12x - 18y + 20)}{\partial y} = 6y - 18$$

$$\text{Gradient } \nabla h(x,y) = [4x - 12, 6y - 18]$$

Finding critical point by equating with 0:

$$\Rightarrow 4x - 12 = 0 \Rightarrow x = 3$$

$$\Rightarrow 6y - 18 = 0 \Rightarrow y = 3$$

Evaluating $h(x,y)$ at critical points & end points:

$$\text{At } (3,3): h(3,3) = 2 \cdot 3^2 + 3 \cdot 3^2 - 12 \cdot 3 - 18 \cdot 3 + 20 = -25$$

$$\text{At } (1,0): h(1,0) = 2 \cdot 1^2 + 3 \cdot 0^2 - 12 \cdot 1 - 18 \cdot 0 + 20 = 10$$

$$\text{At } (1,4): h(1,4) = 2 \cdot 1^2 + 3 \cdot 4^2 - 12 \cdot 1 - 18 \cdot 4 + 20 = -14$$

$$\text{At } (5,0): h(5,0) = 2 \cdot 5^2 + 3 \cdot 0^2 - 12 \cdot 5 - 18 \cdot 0 + 20 = 10$$

$$\text{At } (5,4): h(5,4) = 2 \cdot 5^2 + 3 \cdot 4^2 - 12 \cdot 5 - 18 \cdot 4 + 20 = -14$$

$$\text{Maximum of } h(x,y) = 10 \text{ at } (1,0) \text{ and } (5,0)$$

$$\text{Minimum of } h(x,y) = -25 \text{ at } (3,3)$$

5. Solution given

Example problem on Jacobian:

Consider a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by:

$$f(x, y) = \begin{bmatrix} x^2 + y \\ x - y^2 \end{bmatrix}$$

Find the Jacobian.

→ Computing partial derivatives of each component function:

1. For $f_1(x, y) = x^2 + y$:

$$\frac{\partial f_1}{\partial x} = \frac{\partial (x^2 + y)}{\partial x} = 2x$$

$$\frac{\partial f_1}{\partial y} = \frac{\partial (x^2 + y)}{\partial y} = 1$$

2. For $f_2(x, y) = x - y^2$

$$\frac{\partial f_2}{\partial x} = \frac{\partial (x - y^2)}{\partial x} = 1$$

$$\frac{\partial f_2}{\partial y} = \frac{\partial (x - y^2)}{\partial y} = -2y$$

Thus, The Jacobian ~~matrix~~ is formed as:

$$J_f(x, y) = \begin{bmatrix} 2x & 1 \\ 1 & -2y \end{bmatrix}$$